

Chapter 1.6 Expected Value

Rick Durrett Probability: Theory and Examples

Hu Jingwu

Built from the local Chapter 1 LLMviki flash cards.

Roadmap for 1.6

Learning goal

Treat expectation as integration and deploy inequalities, change of variables, and limit theorems confidently.

1. Expectation as Lebesgue integral.
2. Linearity, monotonicity, and convergence theorems.
3. Markov, Chebyshev, Jensen, and moment inequalities.
4. Change of variables, tail expectations, inclusion-exclusion, and Bonferroni.

Expected Value as Integral

Definition

For a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$,

$$\mathbb{E}X = \int_{\Omega} X d\mathbb{P},$$

provided the positive and negative parts do not both have infinite integral.

Pitfall

$\mathbb{E}X$ is not defined when $\mathbb{E}X^+ = \mathbb{E}X^- = \infty$.

LLMwiki: Expected value as Lebesgue integral; ID: durrett_ch1_expectation_as_integral

Linearity and Monotonicity

Theorem

If X, Y are integrable and $a, b \in \mathbb{R}$, then

$$\mathbb{E}(aX + bY) = a\mathbb{E}X + b\mathbb{E}Y.$$

If $X \leq Y$ a.s., then $\mathbb{E}X \leq \mathbb{E}Y$ whenever the expectations are defined.

Markov and Chebyshev

Markov

If $Y \geq 0$ and $a > 0$, then

$$\mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}Y}{a}.$$

Chebyshev

If $\mathbb{E}X = \mu$ and $\text{var}(X) = \sigma^2$, then

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

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Markov: How to Prove

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Expectation Limit Theorems

Toolkit

Fatou, monotone convergence, dominated convergence, and bounded convergence all transfer to expectations.

- Fatou: nonnegative sequences give lower semicontinuity.
- MCT: $0 \leq X_n \uparrow X$ implies $\mathbb{E}X_n \uparrow \mathbb{E}X$.
- DCT: $X_n \rightarrow X$ and $|X_n| \leq Y$ with $\mathbb{E}Y < \infty$ implies $\mathbb{E}X_n \rightarrow \mathbb{E}X$.

[LLMwiki: Limit theorems for expectation](#); ID: [durrett_ch1_expectation_limit_theorems](#)

Change of Variables Formula

Theorem

If X has law μ and f is measurable, then

$$\mathbb{E}f(X) = \int f(x) \mu(dx),$$

whenever $f \geq 0$ or $f(X)$ is integrable.

LLMwiki: Change of variables formula; ID: durrett_ch1_change_of_variables

Tail Integral Formula

Formula

For $X \geq 0$,

$$\mathbb{E}X = \int_0^\infty \mathbb{P}(X > t) dt.$$

LLMwiki: Tail integral and nonnegative expectations; ID: durrett_ch1_tail_sum_nonnegative

Inclusion-Exclusion

Formula

For events $A_1, \dots, A_n$,

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i) - \sum_{i < j} \mathbb{P}(A_i A_j) + \cdots + (-1)^{n-1} \mathbb{P}\left(\bigcap_i A_i\right).$$

Exercise 1.6.1

Suppose ϕ is strictly convex. Under the assumptions of Jensen's inequality, show that

$$\phi(EX) = E\phi(X)$$

implies $X = EX$ almost surely.

LLMwiki: Equality case in strict Jensen; ID: `exercise_method_strict_jensen_equality`

Exercise 1.6.1: Proof Skeleton

LLMwiki: Equality case in strict Jensen; ID: `exercise_method_strict_jensen_equality`

Exercise 1.6.2

Suppose $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex. Show that

$$E\phi(X_1, \dots, X_n) \geq \phi(EX_1, \dots, EX_n),$$

provided $E|\phi(X_1, \dots, X_n)| < \infty$ and $E|X_i| < \infty$ for all i .

LLMwiki: Jensen's inequality for integrals; ID: durrett_ch1_jensen_integral

Exercise 1.6.2: Proof Skeleton

LLMwiki: Jensen's inequality for integrals; ID: durrett_ch1_jensen_integral

Exercise 1.6.3

Chebyshev's inequality is and is not sharp.

1. Show that if $0 < b \leq a$ are fixed, there is X with $EX^2 = b^2$ for which $P(|X| \geq a) = b^2/a^2$.
2. Show that if $0 < EX^2 < \infty$, then

$$\lim_{a \rightarrow \infty} \frac{a^2 P(|X| \geq a)}{EX^2} = 0.$$

LLMwiki: Two-point distributions as Chebyshev extremizers; ID: exercise_method_two_point_extremizers

Exercise 1.6.3: Proof Skeleton

LLMwiki: Two-point distributions as Chebyshev extremizers; ID: `exercise_method_two_point_extremizers`

Exercise 1.6.4

One-sided Chebyshev bound.

1. Let $a > b > 0$, $0 < p < 1$, and let X satisfy $P(X = a) = p$, $P(X = -b) = 1 - p$. Apply Chebyshev's inequality to $\phi(x) = (x + b)^2$ and conclude that if Y has the same mean and variance as X , then $P(Y \geq a) \leq p$, with equality when $Y = X$.
2. If $EY = 0$, $\text{var}(Y) = \sigma^2$, and $a > 0$, show

$$P(Y \geq a) \leq \frac{\sigma^2}{a^2 + \sigma^2},$$

and show equality is possible.

LLMwiki: Two-point distributions as Chebyshev extremizers; ID: exercise_method_two_point_extremizers

Exercise 1.6.4: Proof Skeleton

LLMwiki: Two-point distributions as Chebyshev extremizers; ID: `exercise_method_two_point_extremizers`

Exercise 1.6.5

Show that the following lower bounds do not exist:

1. If $\varepsilon > 0$,

$$\inf\{P(|X| > \varepsilon) : EX = 0, \text{var}(X) = 1\} = 0.$$

2. If $y \geq 1$ and $\sigma^2 \in (0, \infty)$,

$$\inf\{P(|X| > y) : EX = 1, \text{var}(X) = \sigma^2\} = 0.$$

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Exercise 1.6.5: Proof Skeleton

LLMwiki: Two-point distributions as Chebyshev extremizers; ID: `exercise_method_two_point_extremizers`

Exercise 1.6.6

Let $Y \geq 0$ with $EY^2 < \infty$. Apply Cauchy–Schwarz to $Y\mathbf{1}_{\{Y>0\}}$ and conclude

$$P(Y > 0) \geq \frac{(EY)^2}{EY^2}.$$

LLMwiki: Cauchy-Schwarz lower bound for positive probability; ID: exercise_method_paley_zygmund_basic

Exercise 1.6.6: Proof Skeleton

LLMwiki: Cauchy-Schwarz lower bound for positive probability; ID: `exercise_method_paley_zygmund_basic`

Exercise 1.6.7

Let $\Omega = (0, 1)$ with Lebesgue measure, let $\alpha \in (1, 2)$, and set

$$X_n = n^\alpha \mathbf{1}_{(1/(n+1), 1/n)}.$$

Show $X_n \rightarrow 0$ almost surely and Theorem 1.6.8 applies with $h(x) = x$ and $g(x) = |x|^{2/\alpha}$, but the X_n are not dominated by an integrable random variable.

LLMwiki: [Limit theorems for expectation](#); ID: `durrett_ch1_expectation_limit_theorems`

Exercise 1.6.7: Proof Skeleton

LLMwiki: Limit theorems for expectation; ID: durrett_ch1_expectation_limit_theorems

Exercise 1.6.8

Suppose a probability measure μ has

$$\mu(A) = \int_A f(x) dx$$

for all Borel A . Show that for any g with $g \geq 0$ or $\int |g| d\mu < \infty$,

$$\int g(x) \mu(dx) = \int g(x) f(x) dx.$$

LLMwiki: Change of variables formula; ID: durrett_ch1_change_of_variables

Exercise 1.6.8: Proof Skeleton

LLMwiki: Change of variables formula; ID: durrett_ch1_change_of_variables

Exercise 1.6.9

Let $A_1, \dots, A_n$ be events and $A = \bigcup_{i=1}^n A_i$. Prove the inclusion–exclusion formula.

LLMwiki: Indicator expansion for inclusion-exclusion and Bonferroni; ID:
`exercise_method_inclusion_exclusion_indicators`

Exercise 1.6.9: Proof Skeleton

LLMwiki: Indicator expansion for inclusion-exclusion and Bonferroni; ID:
`exercise_method_inclusion_exclusion_indicators`

Exercise 1.6.10

Prove the Bonferroni inequalities for $A = \bigcup_{i=1}^n A_i$: stopping inclusion–exclusion after an even number of sums gives a lower bound, and stopping after an odd number of sums gives an upper bound.

LLMwiki: Indicator expansion for inclusion-exclusion and Bonferroni; ID:

`exercise_method_inclusion_exclusion_indicators`

Exercise 1.6.10: Proof Skeleton

LLMwiki: Indicator expansion for inclusion-exclusion and Bonferroni; ID:
`exercise_method_inclusion_exclusion_indicators`

Exercise 1.6.11

If $E|X|^k < \infty$, show that for $0 < j < k$,

$$E|X|^j < \infty, \quad E|X|^j \leq (E|X|^k)^{j/k}.$$

LLMwiki: Jensen's inequality for integrals; ID: durrett_ch1_jensen_integral

Exercise 1.6.11: Proof Skeleton

LLMwiki: Jensen's inequality for integrals; ID: durrett_ch1_jensen_integral

Exercise 1.6.12

Apply Jensen's inequality with $\phi(x) = e^x$ and $P(X = \log y_m) = p(m)$ to prove

$$\sum_{m=1}^n p(m)y_m \geq \prod_{m=1}^n y_m^{p(m)}$$

when $\sum_m p(m) = 1$ and $p(m), y_m > 0$.

LLMwiki: [Jensen's inequality for integrals](#); ID: durrett_ch1_jensen_integral

Exercise 1.6.12: Proof Skeleton

LLMwiki: Jensen's inequality for integrals; ID: durrett_ch1_jensen_integral

Exercise 1.6.13

If $EX_1^- < \infty$ and $X_n \uparrow X$, show that

$$EX_n \uparrow EX.$$

LLMwiki: Limit theorems for expectation; ID: durrett_ch1_expectation_limit_theorems

Exercise 1.6.13: Proof Skeleton

LLMwiki: Limit theorems for expectation; ID: durrett_ch1_expectation_limit_theorems

Exercise 1.6.14

Let $X \geq 0$, but do not assume $E(1/X) < \infty$. Show

$$\lim_{y \rightarrow \infty} yE(1/X; X > y) = 0, \quad \lim_{y \downarrow 0} yE(1/X; X > y) = 0.$$

LLMwiki: Tail truncation for expectation limits; ID: `exercise_method_expectation_tail_truncation`

Exercise 1.6.14: Proof Skeleton

LLMwiki: Tail truncation for expectation limits; ID: exercise_method_expectation_tail_truncation

Exercise 1.6.15

If $X_n \geq 0$, show that

$$E \left(\sum_{n=0}^{\infty} X_n \right) = \sum_{n=0}^{\infty} EX_n.$$

LLMwiki: Limit theorems for expectation; ID: durrett_ch1_expectation_limit_theorems

Exercise 1.6.15: Proof Skeleton

LLMwiki: Limit theorems for expectation; ID: durrett_ch1_expectation_limit_theorems

Exercise 1.6.16

If X is integrable and A_n are disjoint sets with union A , show

$$\sum_{n=0}^{\infty} E(X; A_n) = E(X; A),$$

i.e. the sum converges absolutely and has the displayed value.

LLMwiki: [Linearity and monotonicity of expectation](#); ID: durrett_ch1_expectation_linearity

Exercise 1.6.16: Proof Skeleton

LLMwiki: Linearity and monotonicity of expectation; ID: durrett_ch1_expectation_linearity